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Peer Review File

Machine learning for risk stratification in the emergency department (MARS-ED): a randomized controlled trial

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This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors addressed the reviewers' comments well and rephrased the stronger language with more quantitative descriptions. The authors added a 7-day mortality risk assessment to the analysis and discussion, and it may be useful to consider the 7-day mortality risk instead of the 31-day mortality risk score as an avenue for better adoption of this score in clinical practice. Future work should involve a more user-friendly integration of the score into clinical workflow instead of being presented by a research member. Furthermore, there may be better incorporation of the risk score in clinical workflow if it includes a risk "zone", such as infection, or respiratory concern, etc.

Reviewer #2

(Remarks to the Author)
This manuscript represents a rigorous and timely contribution to the evaluation of machine learning–based decision support in the emergency department. The authors have been highly responsive to reviewer feedback, providing substantive clarifications and thoughtful revisions that have strengthened the clarity, methodological rigor, and overall presentation of the work.

In this revision, the authors have effectively addressed reviewer concerns and conveyed to the general readership that, while the model demonstrated good predictive performance, it failed to influence clinician behavior or secondary outcomes and was perceived by clinicians as having low added value. The authors have also made clear that the model was not integrated into clinical workflow, and they thoughtfully reflect on the need to work collaboratively with clinical end-users to revise and optimize the model to enhance clinician trust and perceived value. Their engagement with the review process demonstrates commendable scientific diligence, and the revised version reflects a significant improvement that enhances both readability and impact.

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

<u>Reviewer #1:</u>

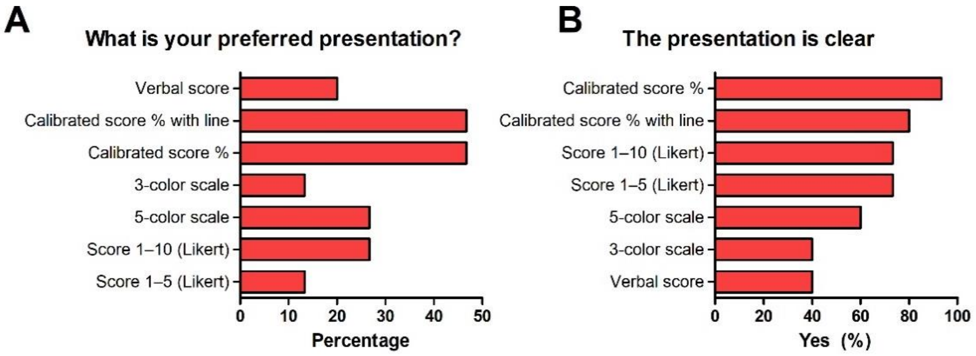
The authors have provided meaningful and satisfactory responses to most of my concerns and questions, but a few significant concerns remain in the following domains:
Thank you. Your remaining concerns are addressed below.

A. The paper presents a randomized implementation of a risk score for the ED, finding adequate predictive accuracy, but no effect on clinical decision-making or clinical outcomes. Clinical course only changed for 1 single patient of over 600 in the intervention arm.
B. With the updated information, I think the main source of information for readers is not necessarily the accuracy of this specific score (the RISK^{INDEX}) for a specific population (there's plenty of papers claiming this); but the fact that this was implemented as part of a clinical trial where the actual impact to clinical practice and clinical outcomes could be assessed. With this lens, given the negative outcome for both clinical outcomes and treatment or evaluation decisions, the paper provides a sort of "negative example" of how NOT to implement a clinical risk score that seems to have adequate predictive accuracy. For readers to extract the most amount of information from it, it is important to detail in the methods and the discussion how the implementation was done, and what was not done.
We agree that the RISK^{INDEX} shows adequate predictive accuracy but had no effect on clinical decision-making or clinical outcomes.
In line with reviewers' helpful suggestions, we now provide a full paragraph with more detailed information on the study preparation and implementation strategy in the Methods section (Online methods, subheading Study preparation, lines 371-382) and the Discussion section (Discussion, lines 170-174 and 199-204).
Moreover, we made several adjustments in the Discussion section to reflect on the implementation strategy, for example (lines 201-204): "*As such, our implementation strategy serves as a valuable negative example—highlighting that even models with strong prognostic accuracy can fail to influence clinical practice when not fully aligned with clinical workflows and decision priorities—while offering important lessons for future research on AI models.*".

C. Hence, the main thing that I still find a significant limitation in the current manuscript is that, while the current manuscript includes additional information about the implementation approach, I still find it an insufficient level of detail in the methods, and very little discussion of the learnings in the discussion section. For example, it provides no information about:
1. what information physicians may have received about the trial beforehand, if any (say, weeks before the trial was started)?
2. At what point in the physicians' workflow (from initial assessment to placing orders to patient admission or discharge) the information about the RISK score was shared. This can significantly affect the conclusions of the study if, for example, information was shared with physicians after being "mentally committed" to a specific course of action and having placed initial orders.
3. We have no information about how the researcher shared information about the score: verbally-only? using visual aids about the score or the accuracy of the model?
We have now added a full paragraph on implementation details including (i) what information the physicians have received about the trial beforehand, (ii) the timing of the presentation of the RISK^{INDEX} and (iii) how the RISK^{INDEX} was presented to the physician (Online methods, subheadings Study preparation and Study intervention and procedure, lines 371-397).
To justify the chosen presentation format, we conducted two projects in which (i) we tested different methods of RISK^{INDEX} presentation (figure below) and (ii) we examined the use of the RISK^{INDEX}

presentation in virtual cases. Based on these findings we decided that a calibrated probability was the preferred option. This information is added to the manuscript (Online methods, subheading, Study preparation, lines 371-382; Extended Data Figure 7).

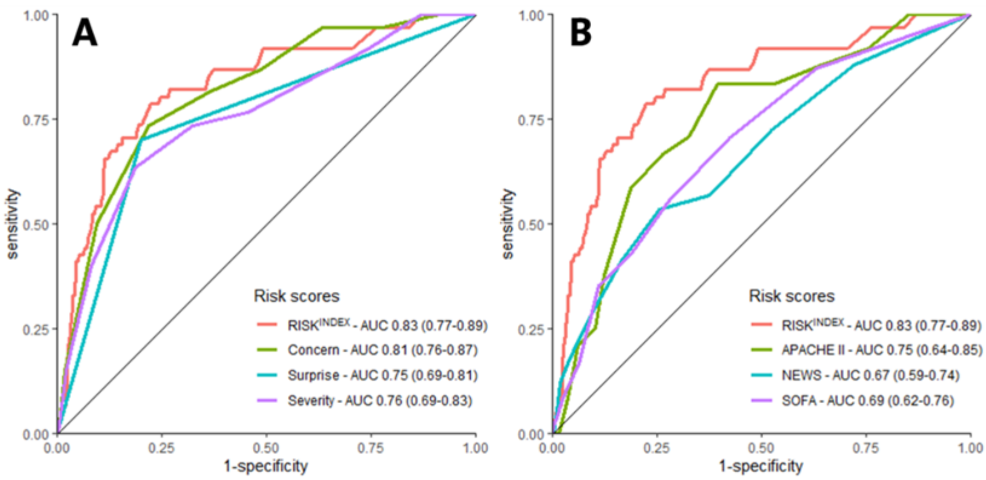
Extended Data Figure 7. Overview of physicians' preferences and clarity ratings for different RISK^{INDEX} presentations. Emergency department physicians (n = 17) rated several RISK^{INDEX} presentation formats for (A) preference and (B) clarity. The calibrated score (with line) was among the most preferred option.



F. Additionally, after having read some of the other reviewers' comments about 7-day mortality being more relevant to ED visits, I would also suggest the authors perform a similar analysis of model-versus-clinician accuracy looking at 7-day mortality. This could potentially help explain the discrepancy between the model predictive accuracy and the physicians' apparent disregard for the predictions. If physicians' predictions more accurately reflect 7-day mortality and this is a more relevant clinical outcome for ED physicians, maybe they were right to ignore the model's predictions.

We thank both reviewers for their suggestion to analyze 7-day mortality to further explore the discrepancy between the RISK^{INDEX}'s predictive accuracy and its clinical impact. The analysis for 7-day mortality and corresponding discussion has now been added to the manuscript (Extended Data Figure 2; Online methods, subsection Statistical analysis, lines 449-450; Results, subheading Primary analysis, lines 113-115; and Discussion, lines 168-169). Briefly, the RISK^{INDEX} shows a similar advantage for 7-day mortality over clinical judgement and traditional risk scores as for 31-day mortality.

Extended Data Figure 2. Receiver-operating-characteristic (ROC) curve analysis of the RISK^{INDEX}, physicians' clinical intuition (concern, surprise and severity questions; A) and traditional clinical prediction tools (APACHE II, NEWS and SOFA; B) for 7-day mortality. The areas under the ROC curves (AUROC) were 0.83 [0.77–0.89], 0.81 [0.76–0.87], 0.75 [0.69–0.81] and 0.76 [0.69–0.83] for the RISK^{INDEX}, concern, surprise and severity questions, respectively (A), and 0.83 [0.77–0.89], 0.75 [0.68–0.82], 0.67 [0.59–0.74] and 0.69 [0.62–0.76] for the RISK^{INDEX}, APACHE II, NEWS and SOFA scores, respectively (B).



Reviewer #2:

Summary of Key Results:

1. The authors revised individual points made by reviewers, however, have not revised the overall message of the manuscript to sufficiently capture the notion that the predictive performance of the model based on AUPRC of 0.33 certainly is better than chance (at 7% for your sample), but references to the model as "superior"/"high" is misleading.

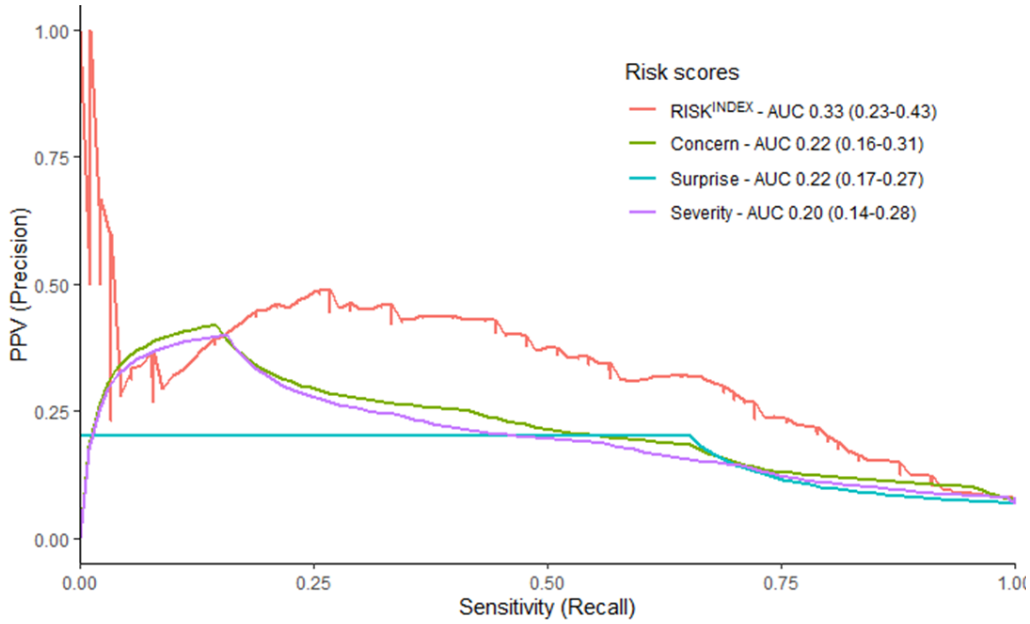
To prevent that readers are misled by the terms "superior/high" we replaced them throughout the manuscript with the term "statistically higher (or equal)".

This AUPRC outperforms the comparators (NEWS APACHE II, and SOFA) by a small/moderate amount, but the comparators are relatively weak in that they are not validated ED EWS scores so that comparison to determine superior/high performance of the ED model is questionable. The AUPRC was not compared to clinician prediction as far as I could find.

We agree that NEWS, SOFA and APACHE II were originally devised for in-hospital or intensive care cohorts. No widely accepted general-purpose risk score currently exists for stratification in the emergency department, and these scores are among the few that have already been evaluated, and clinically used, in ED populations[1-4]. We therefore feel they are the most policy-relevant comparators for our setting. This rationale is clarified in Methods: *"Although originally derived for inpatient and intensive-care cohorts, these three scores are among the few that have been validated, and clinically used, in ED populations and were therefore used as comparators for our study."* (Online methods, subsection Clinical predictions tools, lines 413-416) and, in keeping with the reviewer's suggestions, we now acknowledge this as a study limitation: *"Fourth, although our comparators (NEWS, SOFA and APACHE II) were not originally designed for use in the ED, they are among the few scores validated in ED populations and therefore constitute a potential benchmark"* (Discussion, lines 204-207).

The precision-recall curve comparison with physicians' clinical intuition is demonstrated in the figure below, and is added to the manuscript (Results, subsection Primary analysis, lines 111-113; Extended Data Figure 1).

Extended Data Figure 1. Precision-recall (PR) curve analysis of the $RISK^{INDEX}$ and physician's clinical intuition for 31-day mortality. The area under the precision-recall curves (AUPRC) 0.33 [0.23-0.43], 0.22 [0.16-0.31], 0.22 [0.17-0.27] and 0.20 [0.14-0.28] for the $RISK^{INDEX}$, concern question, surprise question and severity question, respectively.



2. The AUPRC was only mentioned at the end, but as a reviewer appropriately pointed out should be the metric used for an EWS study as opposed to AUROC. While AUROC was performed for the clinician prediction compared to the model, I could not find a reported AUPRC for clinician prediction in order to compare the model AUPRC.
This is now added to the manuscript as Extended Data Figure 1.

3. While the authors acknowledge that this is a negative study in comments to reviewers etc, the abstract and overall summary discussion of findings do not convey to general readership that this study had a model of good predictive performance and failed to influence clinician behavior or secondary outcomes and only had low perceived added value by clinicians. Rather the authors continue to discuss the potential of the model to impact outcomes on a theoretical basis rather than focusing on the negative findings observed. I agree that next steps should include working with clinicians, but that should be to revise the goal and approach to the predictive model, not just how to fit it into workflow.
We have reframed the manuscript to present this as a negative study to the general readership, highlighting key design lessons and acknowledging the importance of user-centered methods in model development. Accordingly, we now explicitly state that the model did not change decision-making or clinical outcomes. Furthermore, we have adjusted descriptions such as "superior/high" to the more nuanced "statistically higher."

B. Originally and significance:
The manuscript should be more clear than current revisions that the model is not integrated into the clinician workflow. Rather it is manually presented by research staff. There were minimal but not sufficient language changes to reflect make sure this is consistency conveyed. For example there remain mentions that the model is automated and other terms that infer it is integrated into the workflow/EHR in an automated way.
We removed terminology related to "automated" in the manuscript and further clarified the presentation of the RISK^{INDEX}: *".., the RISK^{INDEX} was personally presented by a research member to the attending physician, .."* (Online methods, subsection Study intervention, lines 387-389) and *"..., despite extensive physician briefing and direct, in-person presentation of the RISK^{INDEX} predictions."* (Discussion, lines 199-201).

C. Data and Methodology:
1. As mentioned above comparison to models not developed for or validated in the ED setting is a weakness that is not made explicit. In fact references are included for these models but are not for use of these models in the ED setting.
This limitation is now described in our discussion section: *"Fourth, although our comparators (NEWS, SOFA and APACHE II) were not originally designed for emergency department use, they are among the few scores validated, and clinically used, in ED populations and therefore constitute a potential benchmark"* (lines 204-207).

2. unclear justification for interpreting a 2/10 for perceived added value by clinicians as modest which may infer middle to some rather than low.
We agree with reviewer that "modest" might be misinterpreted and therefore changed the word *"modest"* to *"low"* throughout the manuscript (lines 43, 139, 158, 174).

D. Statistics - see comments above for AUPRC
This has now been added to the manuscript as Extended Data Figure 1.

E. Conclusions - Much could be learned from this as a negative study, but reframing is needed given continued focus on the superior/high performance of the model. Given clinicians only ranked the model as 2/10 for perceived usefulness and we should listen to clinicians and go back to model building step. This is not a bad thing, it is part of the process and should be acknowledged head on and discussed scientifically. This study is a good example of evidence that we need to explore how to develop models that are clinically relevant and that clinician trust - it is imperative that user-centered design methods are part of the model development process, not just model implementation. The manuscript does not adequately reflect on the need to work with clinical end-users (beyond authors) to revise and optimize the model itself to increase clinician trust and perceived value, but the findings indicate a need to.

We agree that the primary contribution of our study is not the model's discrimination, but the negative impact findings and what they teach us about designing clinically relevant and user-centered models. Accordingly, we have reframed the manuscript to present this as a negative study with clear design lessons and to acknowledge that user-centered methods are important in model development.

Throughout the manuscript we now explicitly state that the model did not change decision-making or clinical outcomes and furthermore we reframed the "superior/high" to a more nuanced "statistically higher". Importantly, we thoroughly revised our discussion to focus on the need to work with clinical end-users in future studies (Discussion, lines 178-191): "*Accordingly, our findings strongly argue for user-centered design across both model development and implementation. First, future studies should retarget the prediction to actionable decisions in the ED (e.g., 7-day adverse outcomes, admission vs discharge, or need for early escalation), co-specified with ED physicians. Second, the conversion of a 0-100 score into more directive, threshold-linked recommendations (e.g., site-calibrated 'rule-out' and 'high-risk' cut-points tied to explicit actions and acceptable miss rates) may be explored. For example, a hospital may define an acceptable rate of 'low-risk' misclassification (e.g. 1%) based on physicians' risk tolerance for adverse events, resulting in a negative predictive value (NPV) of 99%. This NPV can be used to derive the corresponding $RISK^{INDEX}$ rule-out threshold. Third, patient-level explanations and reliability information alongside recommendations can be considered to address counter-intuition and support trust. Last, the $RISK^{INDEX}$ could be integrated earlier in the workflow (e.g. a dynamic electronic health record-embedded display when laboratory results return) to maximize actionability under time pressure. Future studies should undertake these steps in co-design with clinical end-users, such as frontline ED physicians, to maximize potential clinical value.*", which was also emphasized in our final conclusion (Discussion, lines 212-214): *"Future efforts should couple accurate prediction with user-centered and decision-linked interventions, and test these in trials measuring not only predictive gains but also changes in clinician behavior, patient outcomes, and cost-effectiveness."* and Abstract (lines 44-46): *"Future efforts should combine accurate prediction with user-centered model designs co-created with clinicians to ensure relevance, trust, and actionability."*.

References

1 Guan, G., Lee, C. M. Y., Begg, S., Crombie, A. & Mnatzaganian, G. The use of early warning system scores in prehospital and emergency department settings to predict clinical deterioration: A systematic review and meta-analysis. *PLoS One* **17**, e0265559 (2022). https://doi.org/10.1371/journal.pone.0265559

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3 van Dam, P. *et al.* Head-to-head comparison of 19 prediction models for short-term outcome in medical patients in the emergency department: a retrospective study. *Ann Med* **55**, 2290211 (2023). https://doi.org/10.1080/07853890.2023.2290211

4 Xie, F. *et al.* Benchmarking emergency department prediction models with machine learning and public electronic health records. *Sci Data* **9**, 658 (2022). https://doi.org/10.1038/s41597-022-01782-9